

Embedded Software Final PROJECT

Liquid Vending Machine based on the RFID RC522 Module

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1.Project Description: Context and Objectives

Project Description: Context and Objectives

Context

We had to design and develop a fully functional liquid vending machine controlled by an RFID RC522 module.
The machine manages:

- RFID access
- Liquid ordering (type + quantity)
- 120-second countdown timer
- Attempt/order limits
- Many peripherals.

Objectives

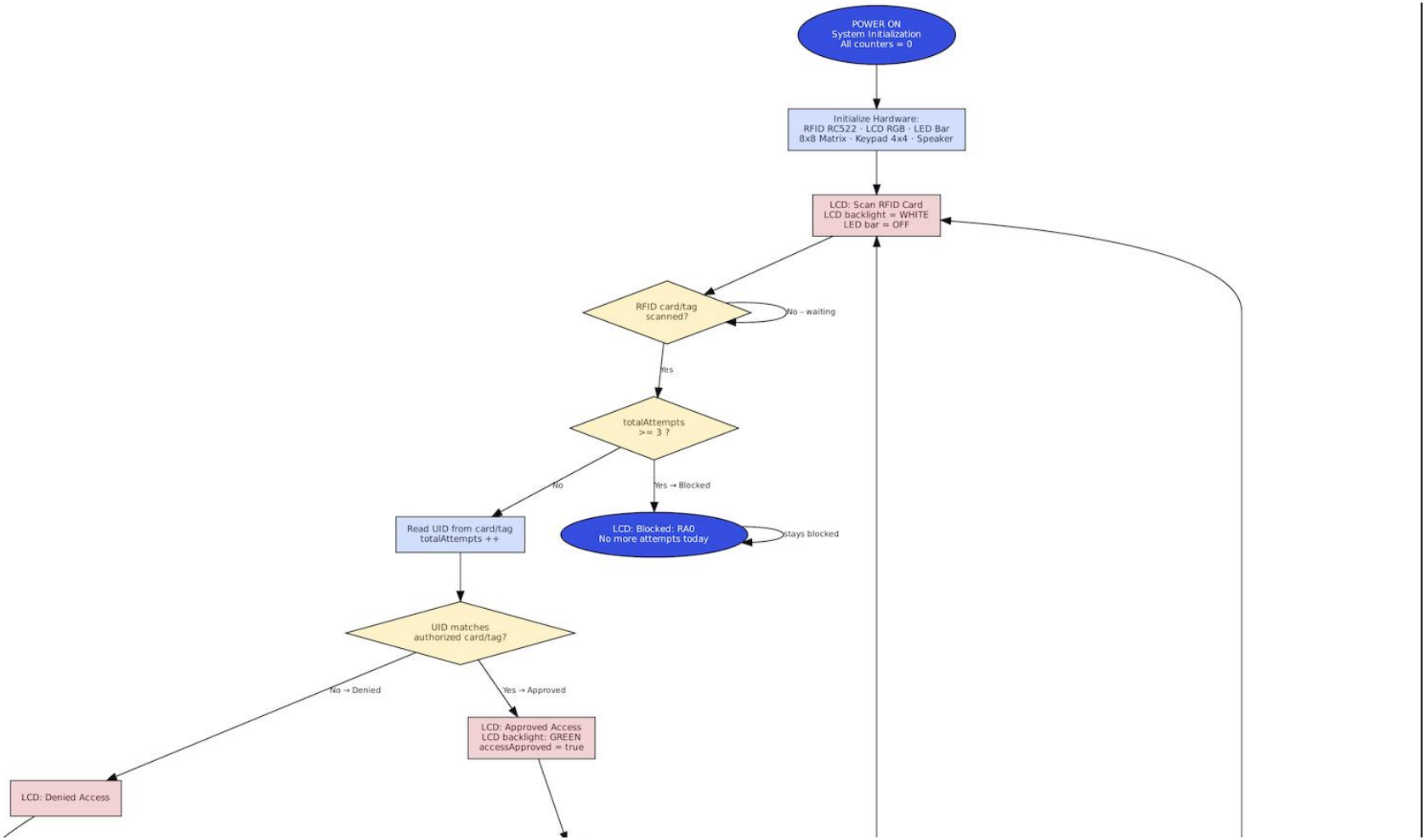
- Part 1 Practical circuit: Wire all components on a breadboard.

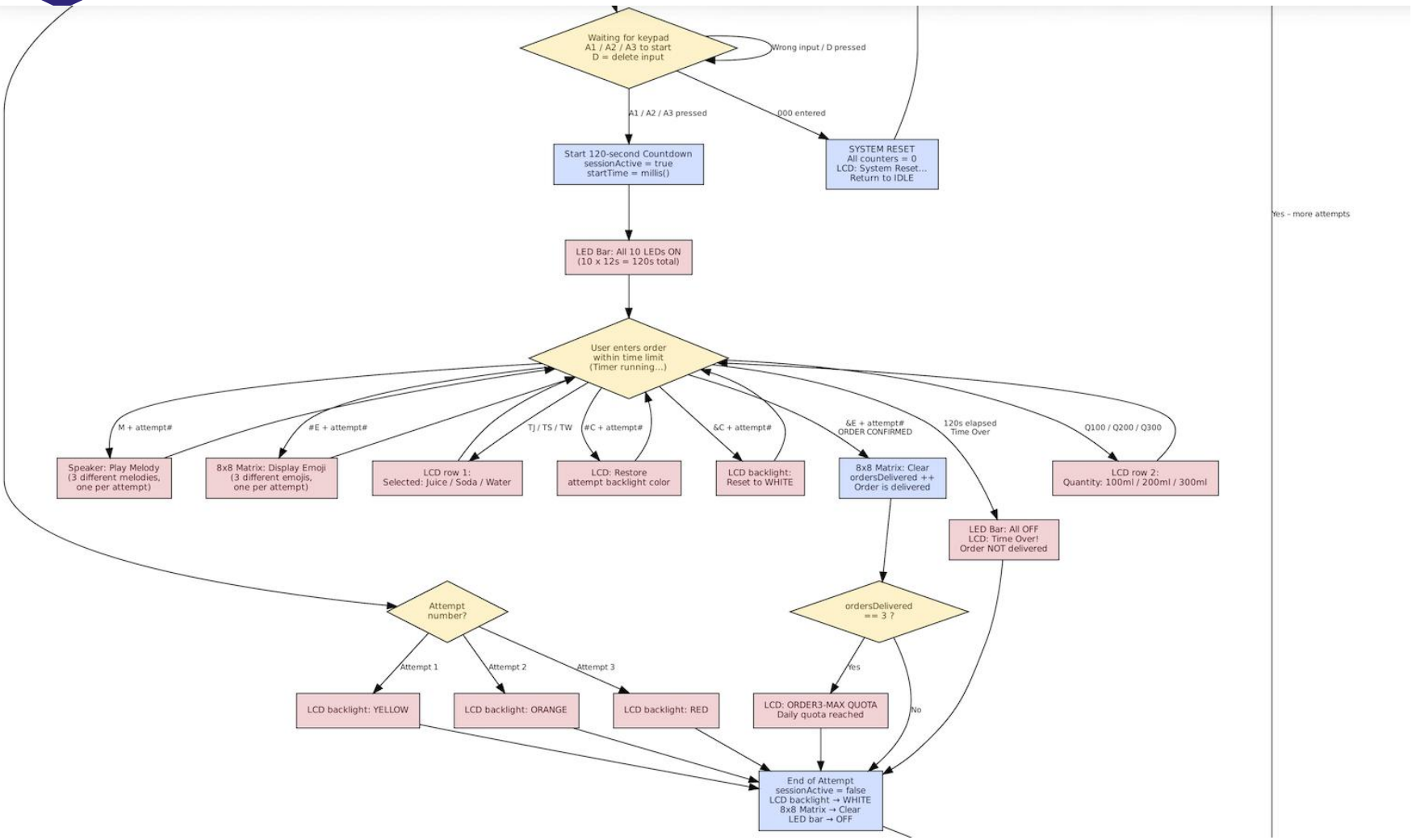
- Part 2 TinkerCAD simulation: Reproduce the system using equivalent available components.

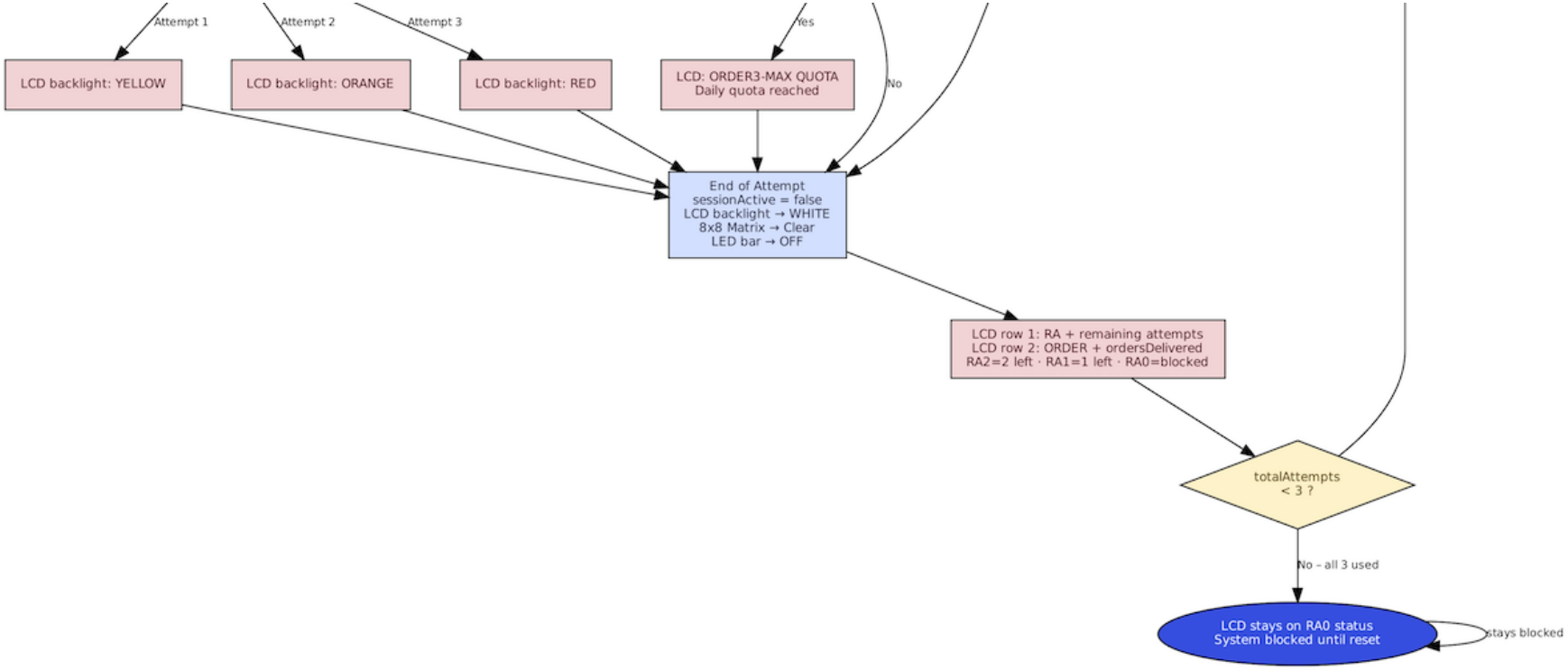
Following all constraints for both parts is a must. Order of procedure etc…

2.Liquid vending machine : The Practical part

2.1. The Flowchart of the practical part (The hardware side)


POWER ON
System Initialization
All counters = 0
Initialize Hardware:
RFID RC522 · LCD RGB · LED Bar
8x8 Matrix · Keypad 4x4 · Speaker
LCD: Scan RFID Card
LCD backlight = WHITE
LED bar = OFF
RFID card/tag scanned?
No – waiting
Yes
totalAttempts >= 3 ?
No
Yes → Blocked
Read UID from card/tag
totalAttempts ++
LCD: Blocked: RA0
No more attempts today
stays blocked
UID matches authorized card/tag?
No → Denied
Yes → Approved
LCD: Approved Access
LCD backlight: GREEN
accessApproved = true
LCD: Denied Access


Waiting for keypad
A1 / A2 / A3 to start
D = delete input
Wrong input / D pressed
A1 / A2 / A3 pressed
000 entered
Start 120-second Countdown
sessionActive = true
startTime = millis()
SYSTEM RESET
All counters = 0
LCD: System Reset...
Return to IDLE
Yes – more attempts
LED Bar: All 10 LEDs ON
(10 x 12s = 120s total)
User enters order
within time limit
(Timer running...)
M + attempt#
#E + attempt#
TJ / TS / TW
#C + attempt#
&C + attempt#
&E + attempt#
ORDER CONFIRMED
120s elapsed
Time Over
Q100 / Q200 / Q300
Speaker: Play Melody
(3 different melodies,
one per attempt)
8x8 Matrix: Display Emoji
(3 different emojis,
one per attempt)
LCD row 1:
Selected: Juice / Soda / Water
LCD: Restore
attempt backlight color
LCD backlight:
Reset to WHITE
8x8 Matrix: Clear
ordersDelivered ++
Order is delivered
LCD row 2:
Quantity: 100ml / 200ml / 300ml
LED Bar: All OFF
LCD: Time Over!
Order NOT delivered
Attempt
number?
ordersDelivered
== 3 ?
Attempt 1
Attempt 2
Attempt 3
Yes
No
LCD backlight: YELLOW
LCD backlight: ORANGE
LCD backlight: RED
LCD: ORDER3-MAX QUOTA
Daily quota reached
End of Attempt
sessionActive = false
LCD backlight → WHITE
8x8 Matrix → Clear
LED bar → OFF


Attempt 1
Attempt 2
Attempt 3
Yes
No
LCD backlight: YELLOW
LCD backlight: ORANGE
LCD backlight: RED
LCD: ORDER3-MAX QUOTA
Daily quota reached
End of Attempt
sessionActive = false
LCD backlight → WHITE
8x8 Matrix → Clear
LED bar → OFF
LCD row 1: RA + remaining attempts
LCD row 2: ORDER + ordersDelivered
RA2=2 left · RA1=1 left · RA0=blocked
totalAttempts
< 3 ?
No – all 3 used
LCD stays on RA0 status
System blocked until reset
stays blocked

- **Component Roles**
- • RFID RC522 (SPI, D9-D13, 3.3V): Reads card/tag UID which grants or denies access
- • 4x4 Keypad (Rows: D2,D6,D7,D8 | Cols: A0-A3): User input
- • Grove LCD RGB Backlight (I2C): shows access status, liquid selection, remaining attempts
- • Grove LED Bar (D4 — Grove Baseshield 5V): Countdown timer. 10 LEDs ON, 1 turns OFF every 12s
- • Grove 8×8 LED Matrix (I2C): Displays specific emoji (#E1 / #E2 / #E3)
- • Speaker + MOSFET BS170 + 330Ω resistor D3 : Plays melody and touch sound

- **Pin Assignments**
- • RFID: VCC=3.3V, RST=D9, GND, MISO=D12, MOSI=D11, SCK=D13, NSS=D10 (IRQ unconnected)
- • Keypad Rows: D2, D6, D7, D8 and Cols: A0(D14), A1(D15), A2(D16), A3(D17)
- • Speaker: D3 through MOSFET BS170 and R1=330Ω
- • Grove Baseshield power switch: 5V

2.2. The Flowchart of the practical part : The Software side

Constraints and Specifications Met

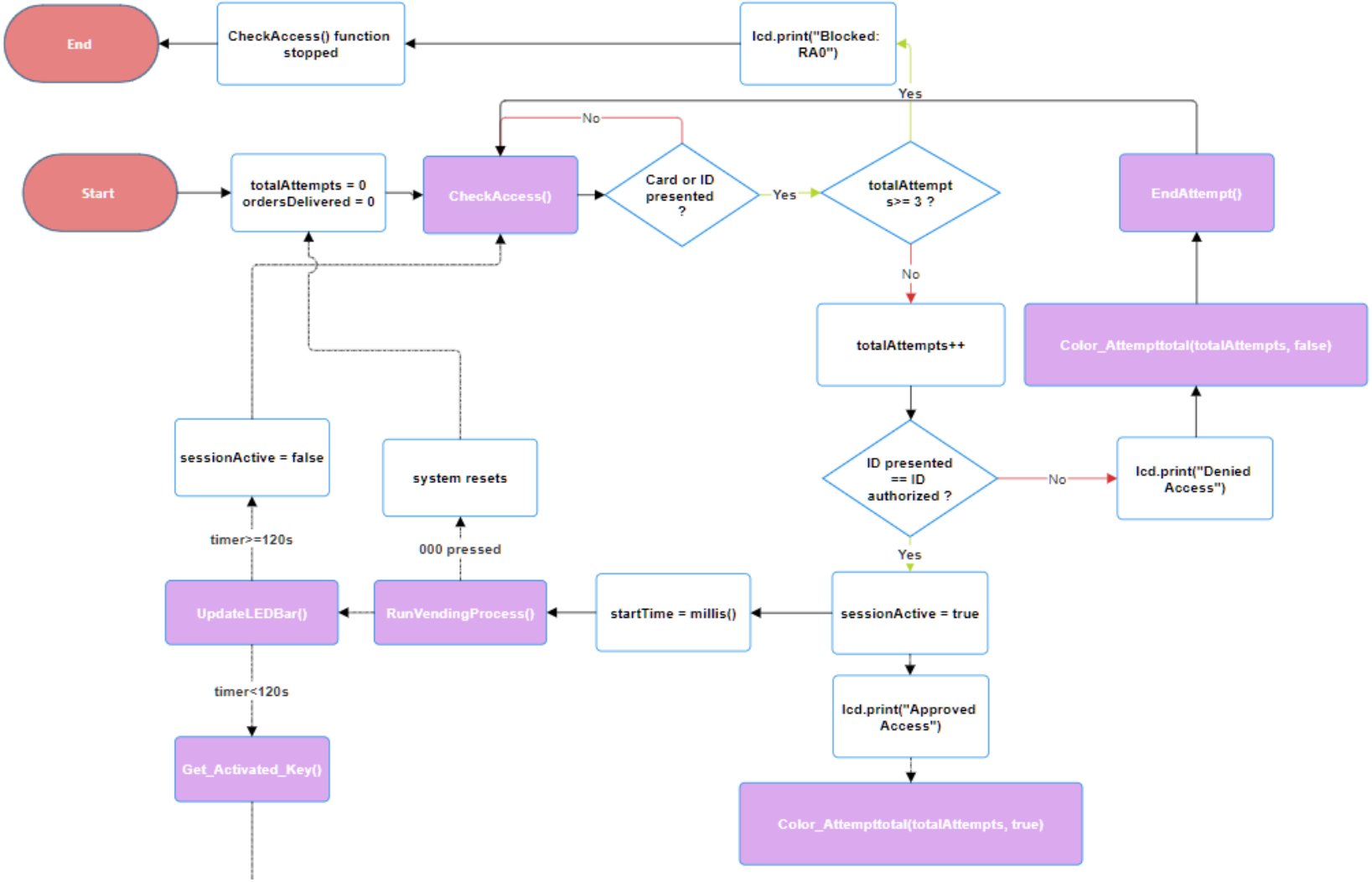
- Constraint 1 — Countdown Timer: 120s per attempt. LED Bar starts at 10 LEDs, 1 turns OFF every 12s.
- Constraint 2 — Attempts/Orders: Max 3 attempts/day, max 3 orders/day.
- Constraint 3 — Keypad Mapping: A=Attempt, T=Type, Q=Quantity, J=Juice, S=Soda, W=Water, M=Melody, E=Emoji, C=Cancel, #/&/D=special.
- Constraint 4 — Emoji Display: 3 distinct custom emojis on 8×8 matrix, one per attempt number.
- Constraint 5 — Melodies: 3 distinct 3-note melodies via speaker (M1: CMario tune | M2: StarWars intro | M3: Pirates of the Caribbean theme).

LCD Backlight Colour — Color_Attempt()

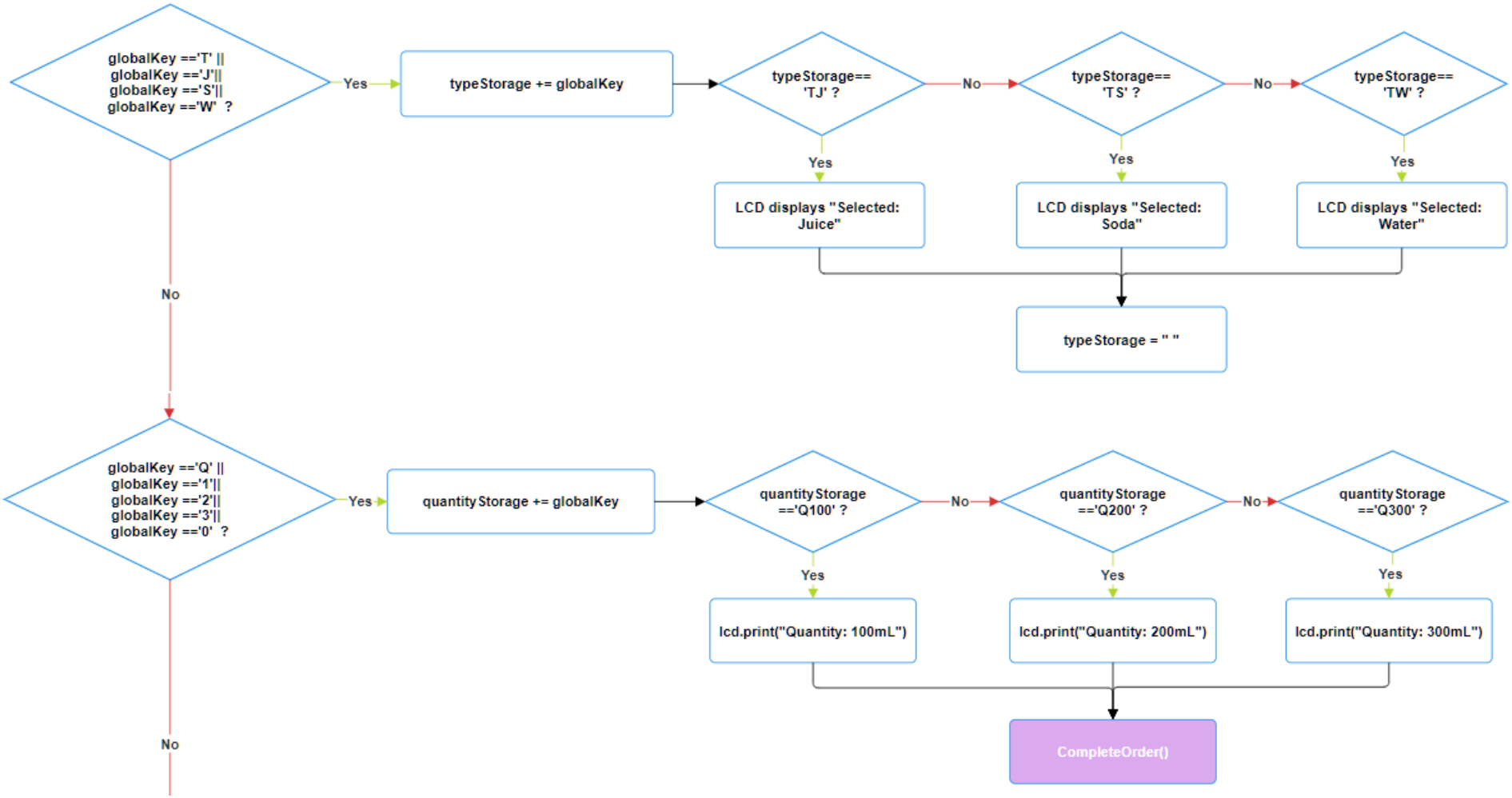
- Access Approved (any attempt): GREEN
- Denied — Attempt 1: YELLOW | Denied — Attempt 2: ORANGE | Denied — Attempt 3: RED

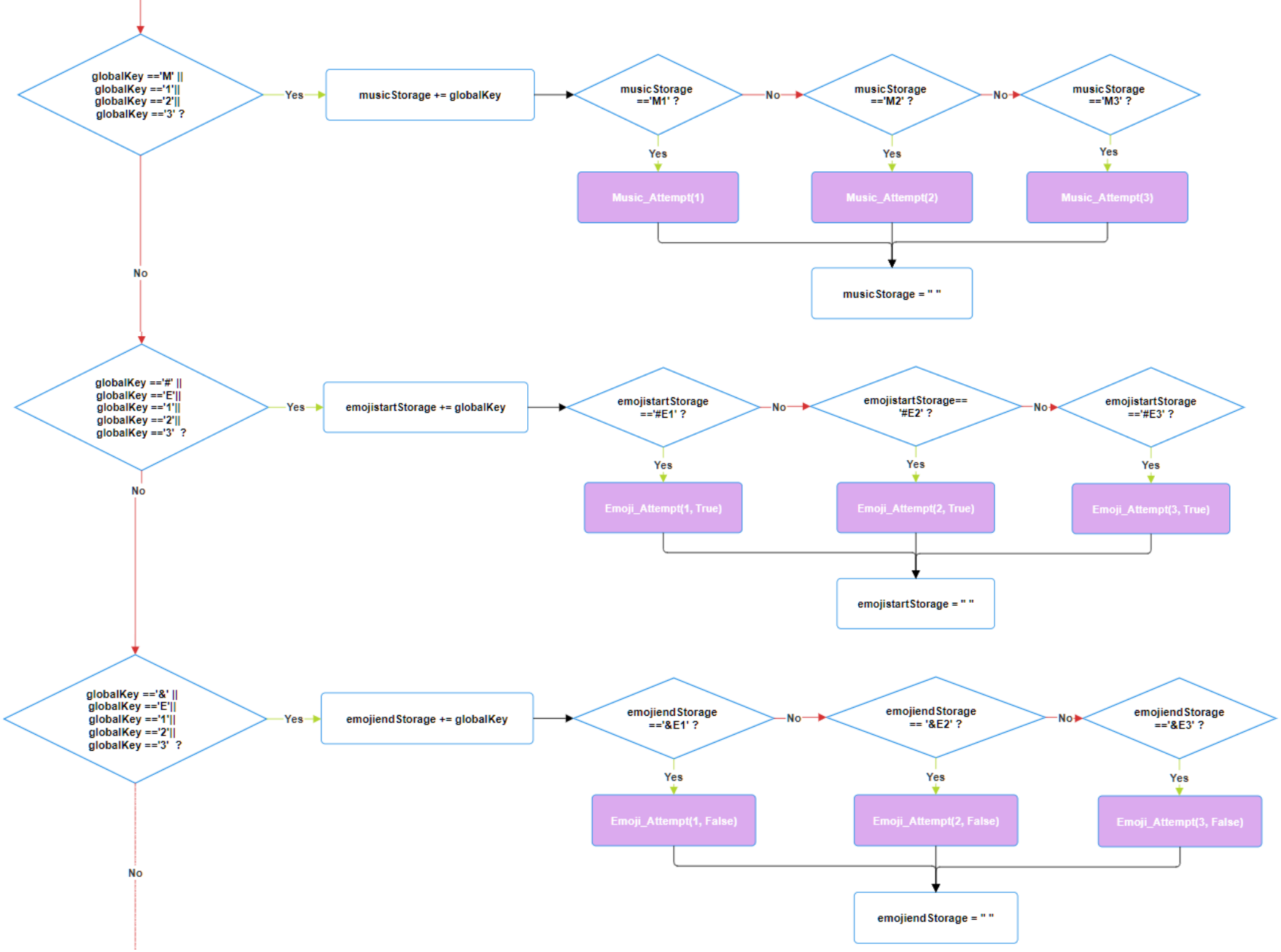
Functions called from void loop()

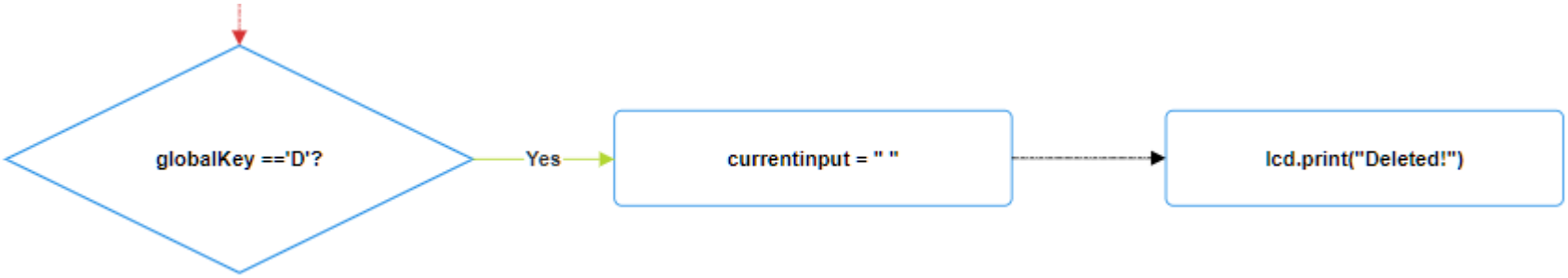
- CheckAccess() — RFID scan and UID verification (authorised card OR tag UID)
- Get_activated_key() — Keypad polling and multi-character input parsing (e.g. "TJ", "Q200")
- RunVendingProcess() — Order logic: liquid type, quantity, LCD updates, order/attempt counters
- UpdateLEDBar() — Reads elapsed time, sets LED Bar level every 12 seconds
- Color_Attempt(), Music_Attempt(), Emoji_Attempt() — Called at appropriate trigger points

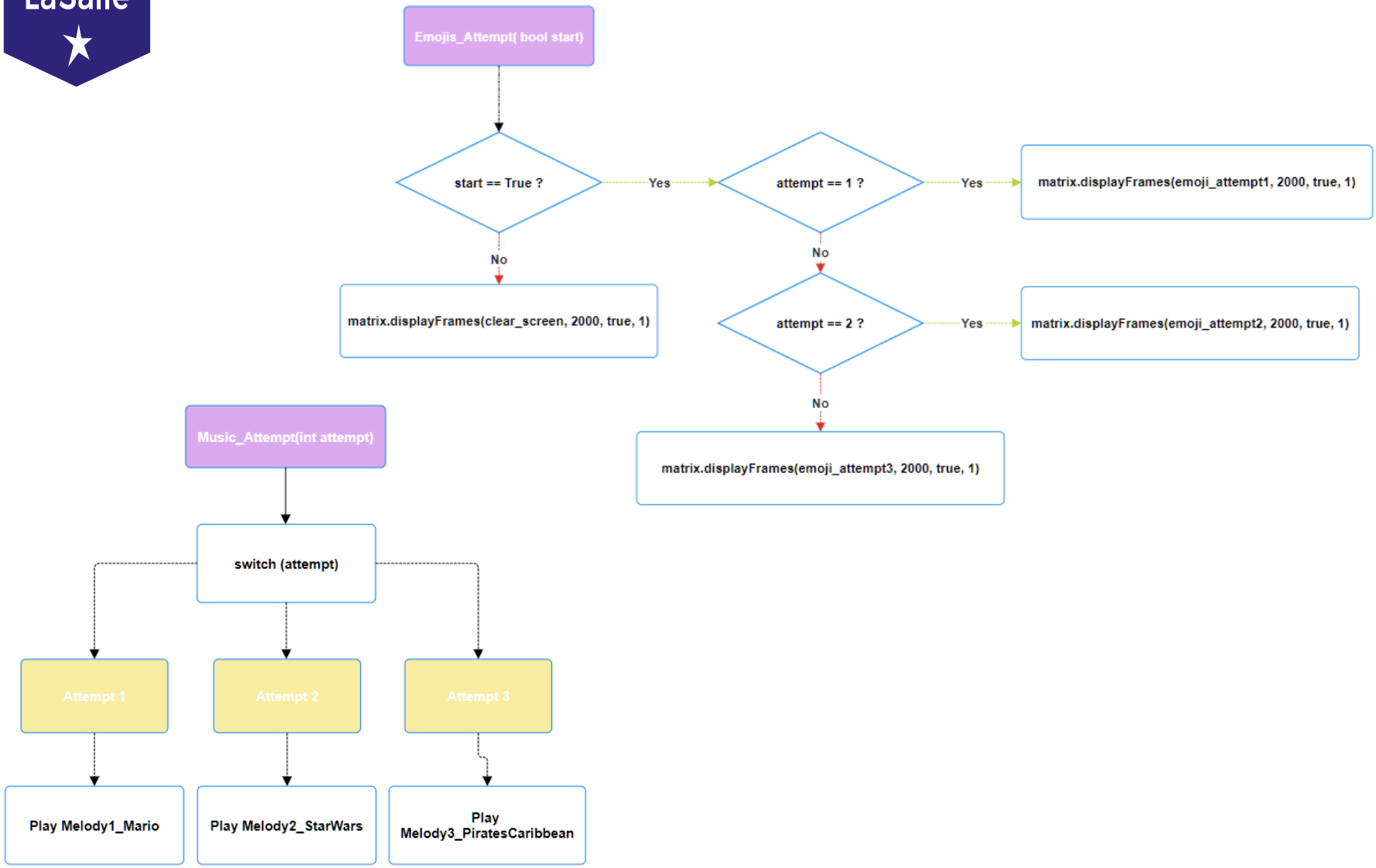

End
CheckAccess() function stopped
lcd.print("Blocked: RA0")
Yes
No
Start
totalAttempts = 0
ordersDelivered = 0
CheckAccess()
Card or ID presented ?
Yes
totalAttempts >= 3 ?
EndAttempt()
No
totalAttempts++
Color_Attempttotal(totalAttempts, false)
sessionActive = false
system resets
ID presented == ID authorized ?
No
lcd.print("Denied Access")
timer>=120s
000 pressed
Yes
UpdateLEDBar()
RunVendingProcess()
startTime = millis()
sessionActive = true
timer<120s
lcd.print("Approved Access")
Get_Activated_Key()
Color_Attempttotal(totalAttempts, true)

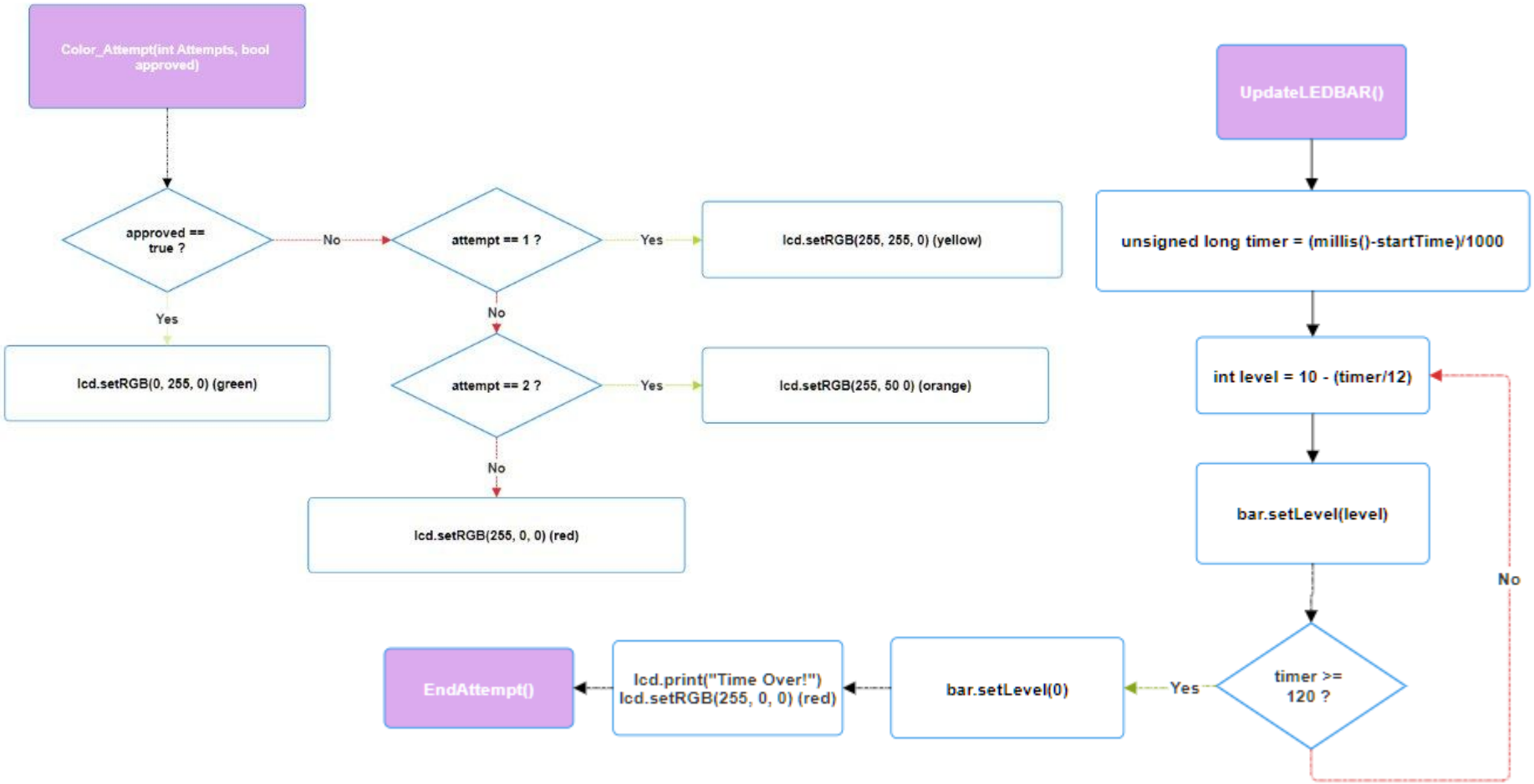
The Get_Activated_Key() function

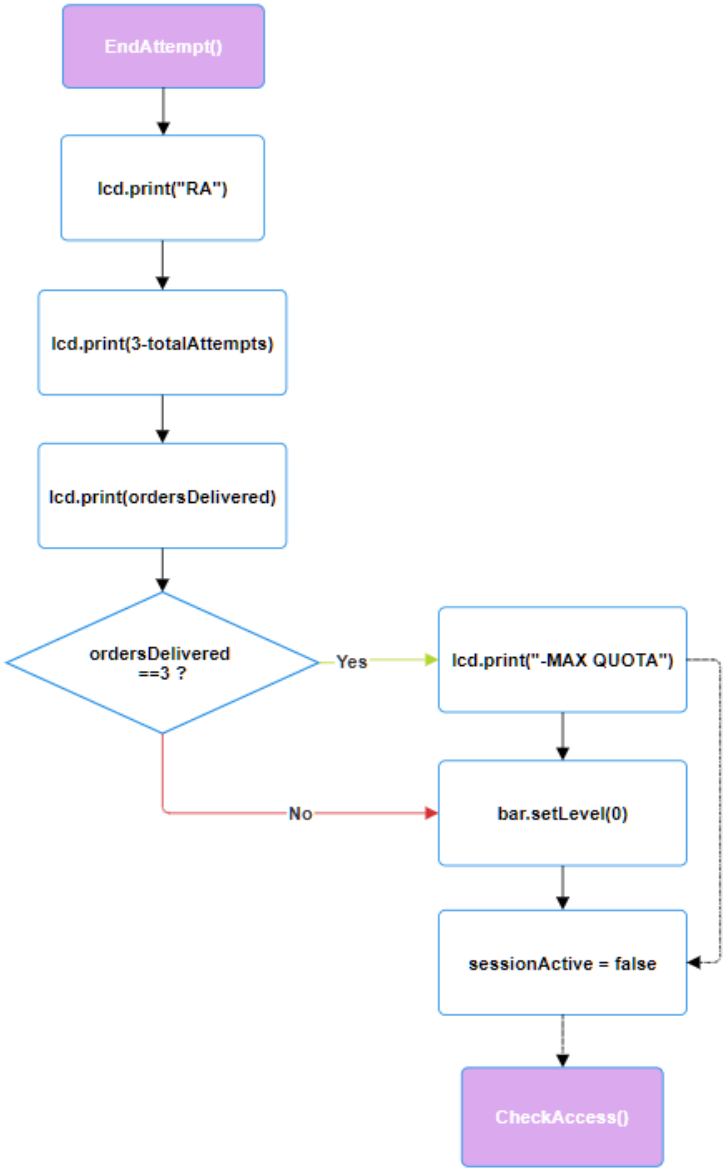


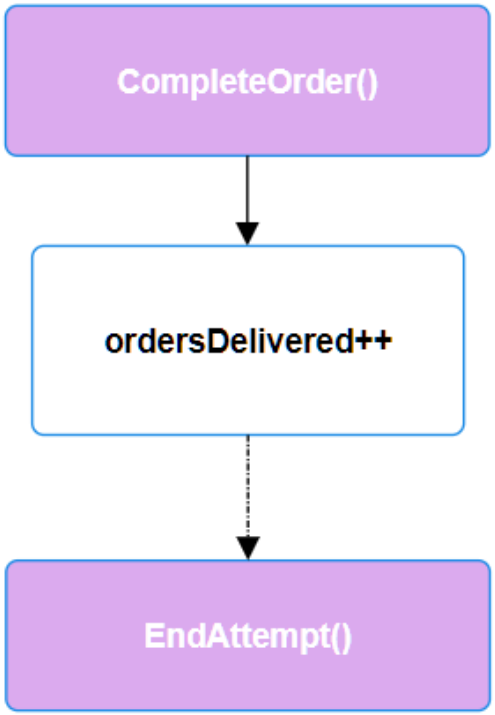

globalKey =='M' ||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
musicStorage += globalKey
musicStorage =='M1' ?
No
musicStorage =='M2' ?
No
musicStorage =='M3' ?
Yes
Yes
Yes
Music_Attempt(1)
Music_Attempt(2)
Music_Attempt(3)
musicStorage = " "
No
globalKey =='#' ||
globalKey =='E'||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
emojistartStorage += globalKey
emojistartStorage =='#E1' ?
No
emojistartStorage== '#E2' ?
No
emojistartStorage =='#E3' ?
Yes
Yes
Yes
Emoji_Attempt(1, True)
Emoji_Attempt(2, True)
Emoji_Attempt(3, True)
emojistartStorage = " "
No
globalKey =='&' ||
globalKey =='E'||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
emojiendStorage += globalKey
emojiendStorage =='&E1' ?
No
emojiendStorage == '&E2' ?
No
emojiendStorage =='&E3' ?
Yes
Yes
Yes
Emoji_Attempt(1, False)
Emoji_Attempt(2, False)
Emoji_Attempt(3, False)
emojiendStorage = " "
No


globalKey =='D'?
Yes
currentinput = " "
lcd.print("Deleted!")

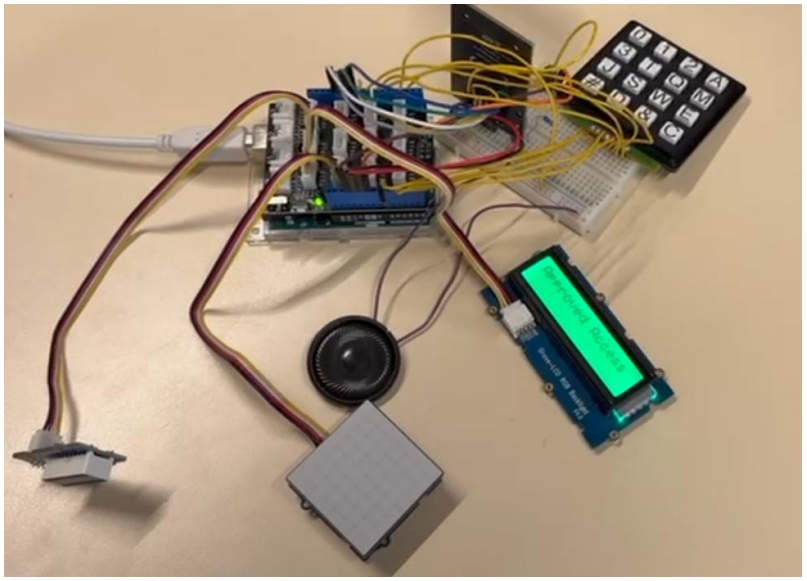

Emojis_Attempt(bool start)
start == True ?
Yes
attempt == 1 ?
Yes
matrix.displayFrames(emoji_attempt1, 2000, true, 1)
No
matrix.displayFrames(clear_screen, 2000, true, 1)
No
attempt == 2 ?
Yes
matrix.displayFrames(emoji_attempt2, 2000, true, 1)
No
matrix.displayFrames(emoji_attempt3, 2000, true, 1)
Music_Attempt(int attempt)
switch (attempt)
Attempt 1
Attempt 2
Attempt 3
Play Melody1_Mario
Play Melody2_StarWars
Play Melody3_PiratesCaribbean


Color_Attempt(int Attempts, bool approved)
approved == true ?
No
attempt == 1 ?
Yes
lcd.setRGB(255, 255, 0) (yellow)
Yes
lcd.setRGB(0, 255, 0) (green)
No
attempt == 2 ?
Yes
lcd.setRGB(255, 50 0) (orange)
No
lcd.setRGB(255, 0, 0) (red)
UpdateLEDBAR()
unsigned long timer = (millis()-startTime)/1000
int level = 10 - (timer/12)
bar.setLevel(level)
timer >= 120 ?
Yes
No
bar.setLevel(0)
lcd.print("Time Over!")
lcd.setRGB(255, 0, 0) (red)
EndAttempt()


EndAttempt()
lcd.print("RA")
lcd.print(3-totalAttempts)
lcd.print(ordersDelivered)
ordersDelivered ==3 ?
Yes
No
lcd.print("-MAX QUOTA")
bar.setLevel(0)
sessionActive = false
CheckAccess()


CompleteOrder()
ordersDelivered++
EndAttempt()

2.3. Obtained Results



Practical circuit on breadboard



LCD "Approved Access" — GREEN backlight

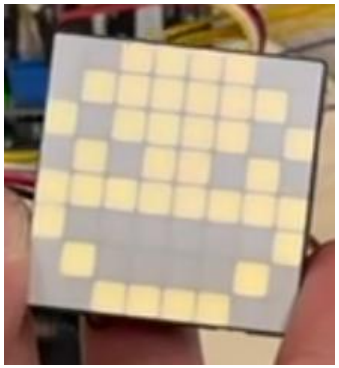
LCD "Denied Access" — YELLOW / ORANGE / RED backlight

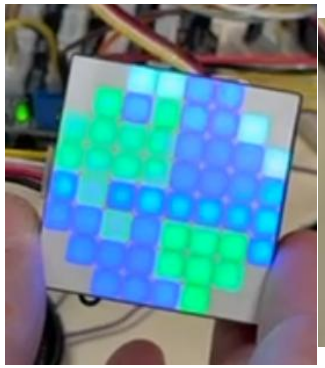


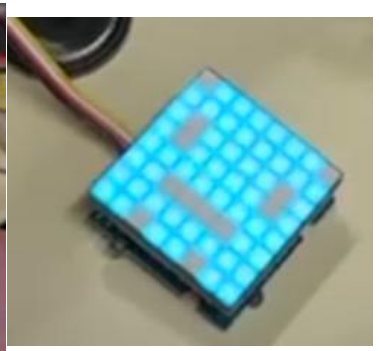












8×8 Matrix showing Emoji for Attempt 1, 2, and 3

LED Bar countdown timer in progress





- *LCD showing "RA2" / "ORDER1" at end of attempt*

LCD showing "Selected: Juice" / "Quantity: 100 ml"

3. Liquid vending machine : the Tinkercad part

3.1. The Flowchart of the tinkercad part (The hardware side)

Objective

Our objective was to simulate a full vending machine software and hardware on tinkercad using the equivalent components and having to improvise for missing components.

Materials Used and Justification

• Push Button that replaces the RFID RC522: the RFID module not available in Tinkercad, so we use a push button to simulate the card or tag being presented

• No denied access function

• I2C LCD + External RGB LED → replaces Grove LCD RGB: Grove LCD library unavailable.

• 1 NeoPixel 10x1 band (+ "Protection" 330 Ohm Resistor) → replaces the Grove LED Bar: the Grove LED Bar library unavailable in TinkerCAD, so each bar will be symbolized as one LED/Pixel on the band.

• 8 NeoPixel 8x1 bands (+ "Protection" 330 Ohm Resistor)→ to replace the 8×8 Grove Matrix: closest available light-pattern component. Even though the current need will be far greater than it would be in real life with the Grove matrix.

• Piezo Buzzer and NPN Transistor → same as practical (available in TinkerCAD). We use a Transistor to boost the signal.

• 4×4 Keypad → identical key mapping as practical version

TinkerCAD Circuit Link:

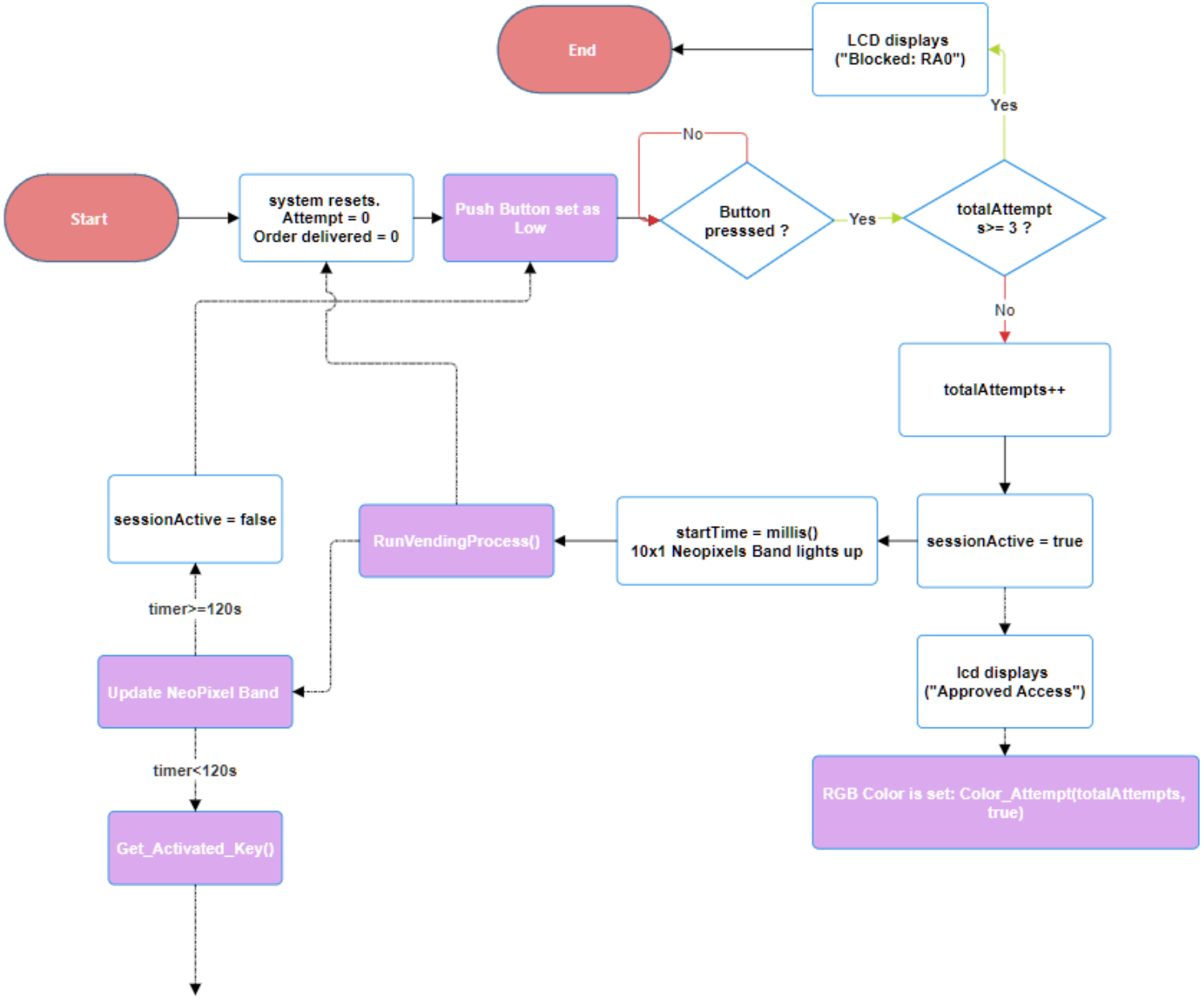
https://www.tinkercad.com/things/inMIphZpRnw-copy-of-embedded-software-project?sharecode=Uc8Cici55s33sYXXtKLnGC1PKduyrO35I9lv0QAza94

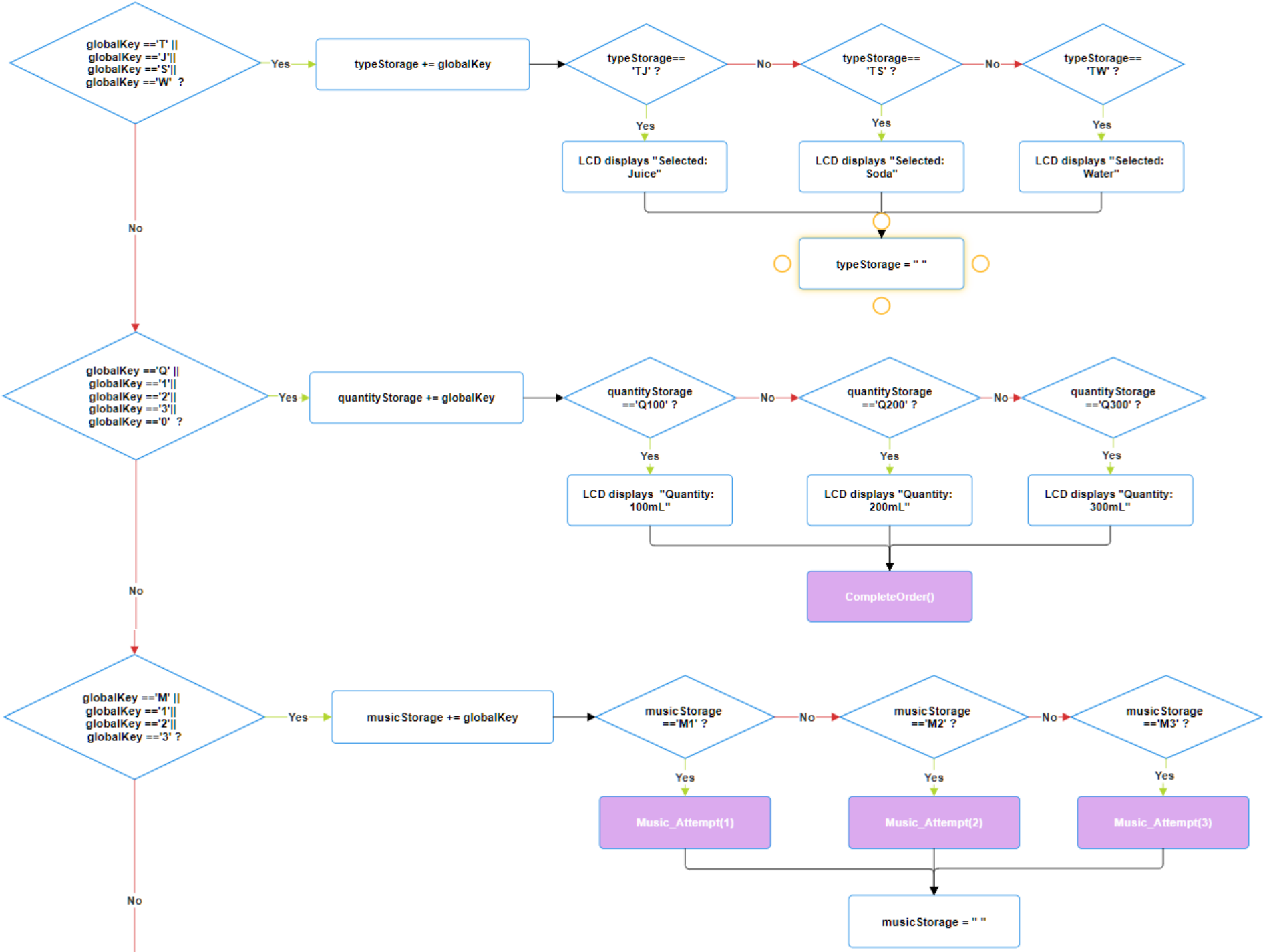
Wokwi Circuit Link:

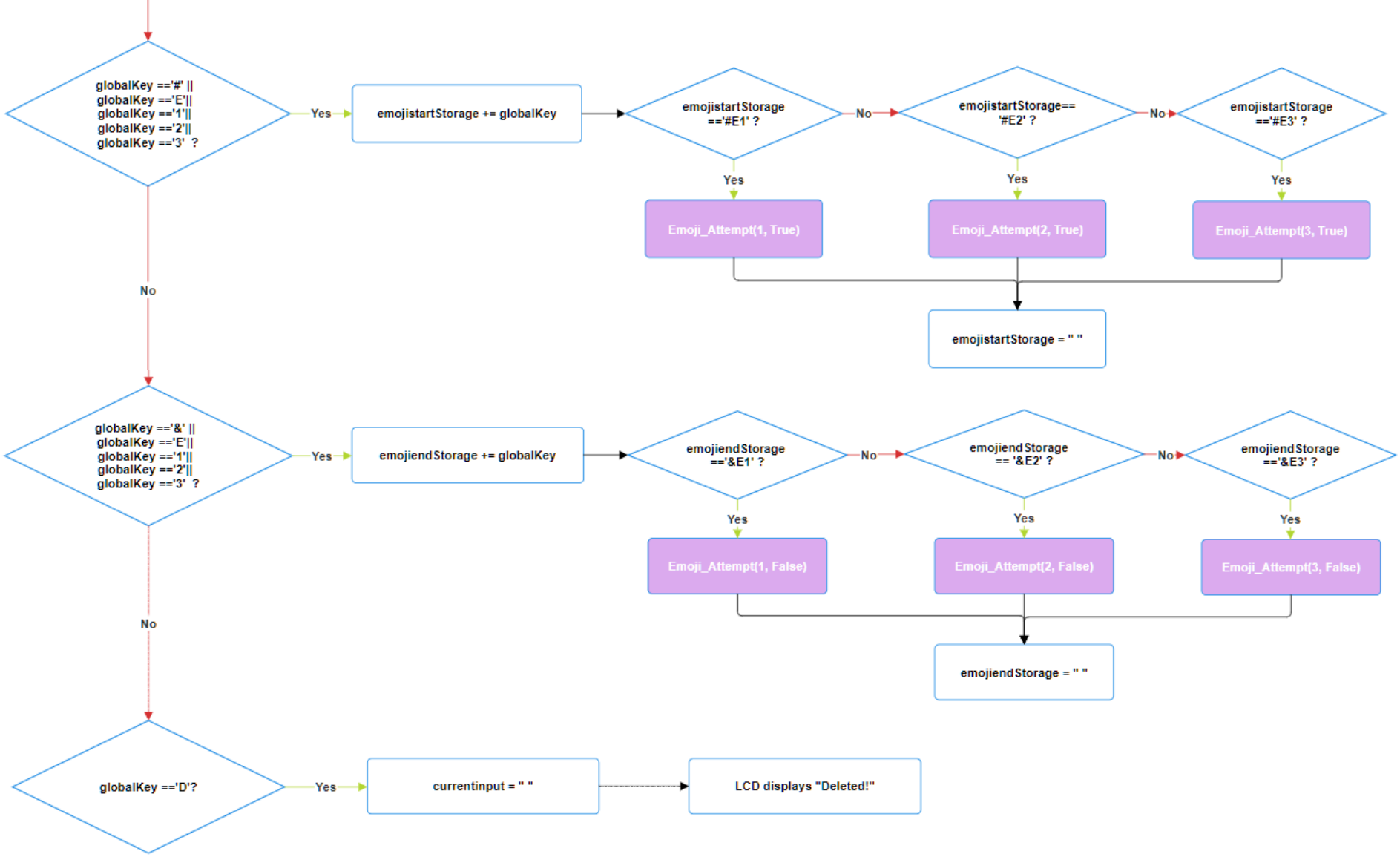
https://wokwi.com/projects/463945473171435 53

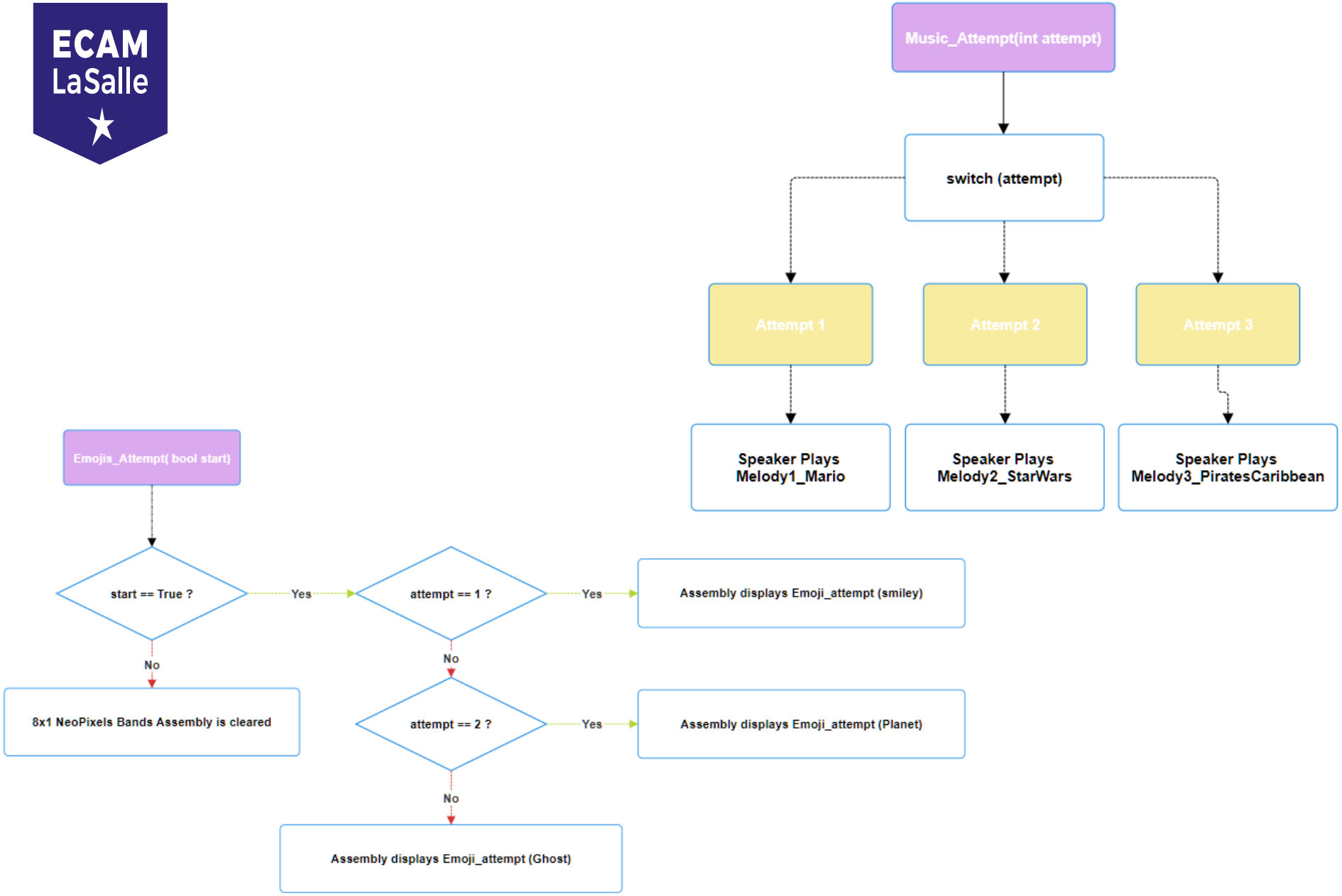
Pin Assignments

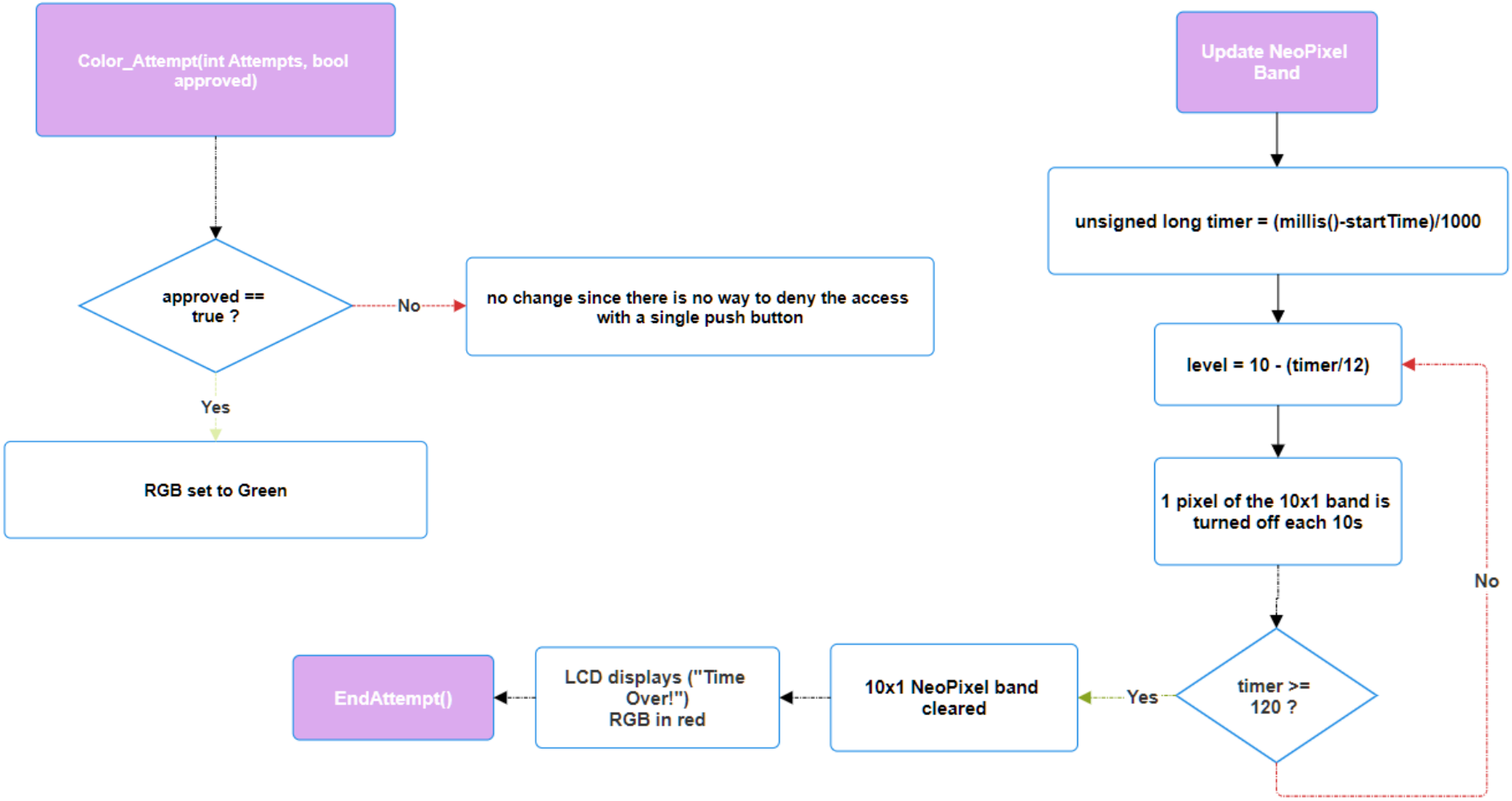
- Button: D10 | RGB LED: D13 (R), D12 (B), D11 (G-PWM) | Buzzer + NPN transistor: D3
- I2C LCD: A4 (SDA), A5 (SCL) | 10x1 Neopixels Band: D4 | 8x(8x1) Neopixel Band Assembly: D9
- 4x4 Keypad: Row Pins (D2, D6, D7, D8) Columns Pins (A0, A1, A2, A3)

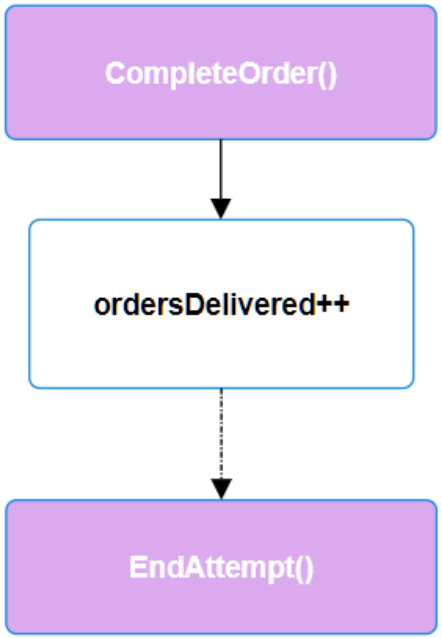

End
LCD displays ("Blocked: RA0")
Yes
No
Start
system resets. Attempt = 0 Order delivered = 0
Push Button set as Low
Button presssed ?
Yes
totalAttempt s>= 3 ?
No
totalAttempts++
sessionActive = false
RunVendingProcess()
startTime = millis() 10x1 Neopixels Band lights up
sessionActive = true
timer>=120s
Update NeoPixel Band
lcd displays ("Approved Access")
timer<120s
RGB Color is set: Color_Attempt(totalAttempts, true)
Get_Activated_Key()

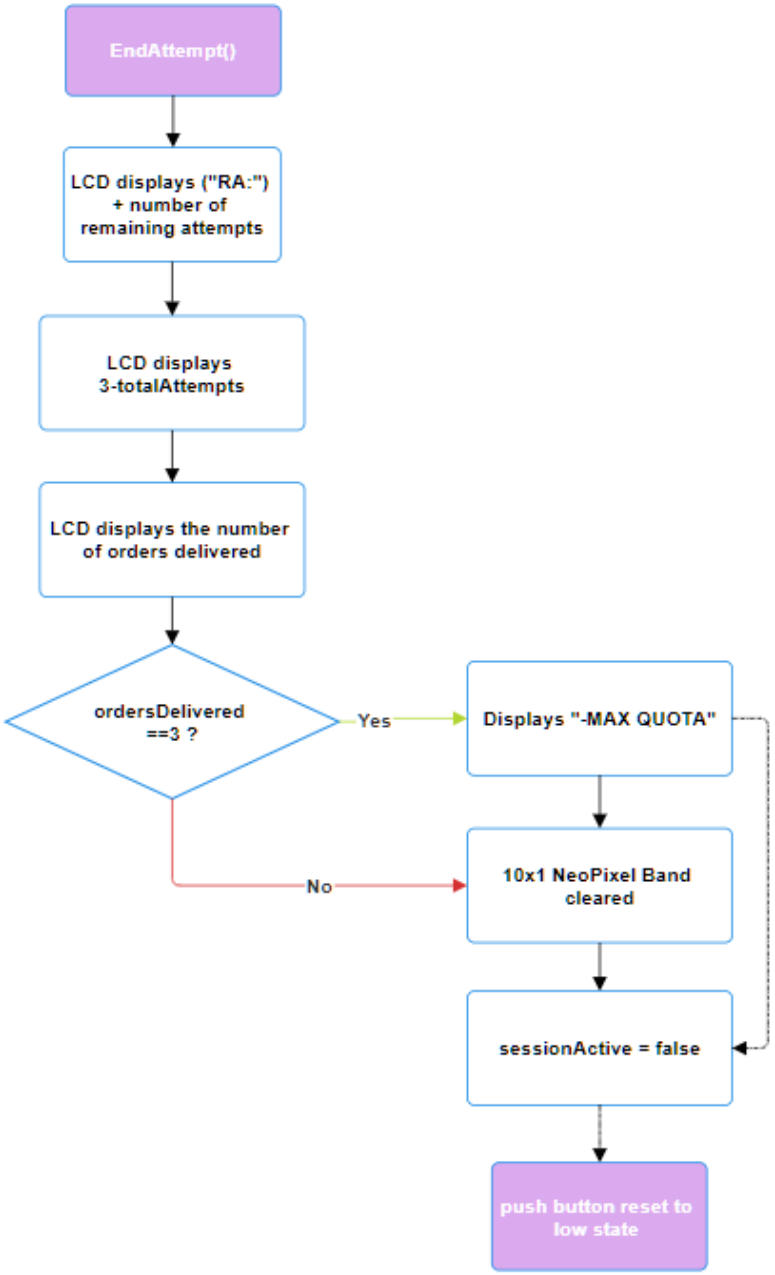

globalKey =='T' ||
globalKey =='J'||
globalKey =='S'||
globalKey =='W' ?
Yes
typeStorage += globalKey
typeStorage== 'TJ' ?
No
typeStorage== 'TS' ?
No
typeStorage== 'TW' ?
Yes
LCD displays "Selected: Juice"
Yes
LCD displays "Selected: Soda"
Yes
LCD displays "Selected: Water"
typeStorage = " "
No
globalKey =='Q' ||
globalKey =='1'||
globalKey =='2'||
globalKey =='3'||
globalKey =='0' ?
Yes
quantityStorage += globalKey
quantityStorage ==Q100' ?
No
quantityStorage ==Q200' ?
No
quantityStorage ==Q300' ?
Yes
LCD displays "Quantity: 100mL"
Yes
LCD displays "Quantity: 200mL"
Yes
LCD displays "Quantity: 300mL"
CompleteOrder()
No
globalKey =='M' ||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
musicStorage += globalKey
musicStorage ==M1' ?
No
musicStorage ==M2' ?
No
musicStorage ==M3' ?
Yes
Music_Attempt(1)
Yes
Music_Attempt(2)
Yes
Music_Attempt(3)
musicStorage = " "
No


globalKey =='#' ||
globalKey =='E'||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
emojistartStorage += globalKey
emojistartStorage ='#E1' ?
No
emojistartStorage== '#E2' ?
No
emojistartStorage ='#E3' ?
Yes
Yes
Yes
Emoji_Attempt(1, True)
Emoji_Attempt(2, True)
Emoji_Attempt(3, True)
emojistartStorage = " "
No
globalKey =='&' ||
globalKey =='E'||
globalKey =='1'||
globalKey =='2'||
globalKey =='3' ?
Yes
emojiendStorage += globalKey
emojiendStorage ='&E1' ?
No
emojiendStorage == '&E2' ?
No
emojiendStorage ='&E3' ?
Yes
Yes
Yes
Emoji_Attempt(1, False)
Emoji_Attempt(2, False)
Emoji_Attempt(3, False)
emojiendStorage = " "
No
globalKey =='D'?
Yes
currentinput = " "
LCD displays "Deleted!"

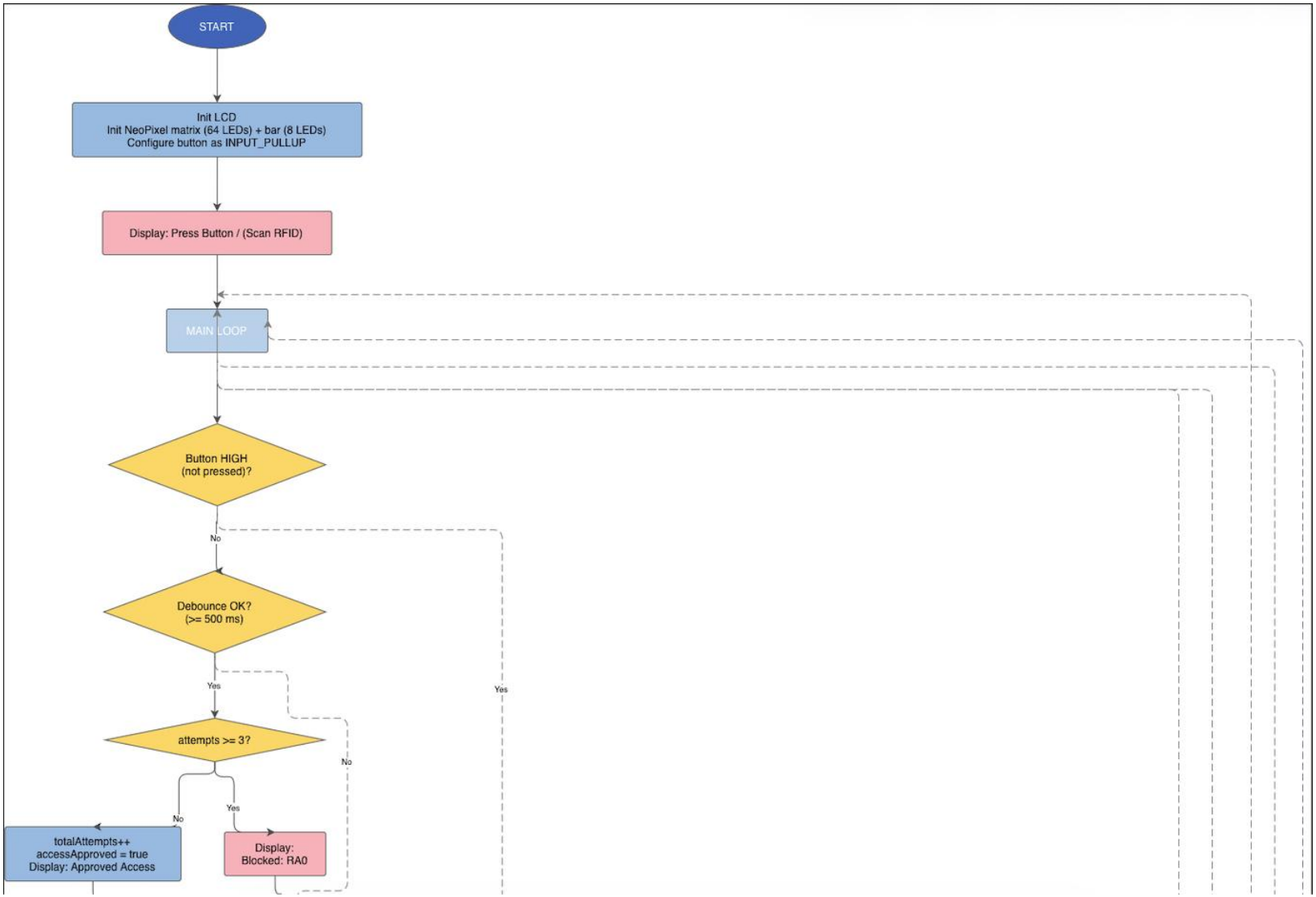

Music_Attempt(int attempt)
switch (attempt)
Attempt 1
Attempt 2
Attempt 3
Speaker Plays Melody1_Mario
Speaker Plays Melody2_StarWars
Speaker Plays Melody3_PiratesCaribbean
Emojis_Attempt(bool start)
start == True ?
Yes
No
8x1 NeoPixels Bands Assembly is cleared
attempt == 1 ?
Yes
Assembly displays Emoji_attempt (smiley)
No
attempt == 2 ?
Yes
Assembly displays Emoji_attempt (Planet)
No
Assembly displays Emoji_attempt (Ghost)

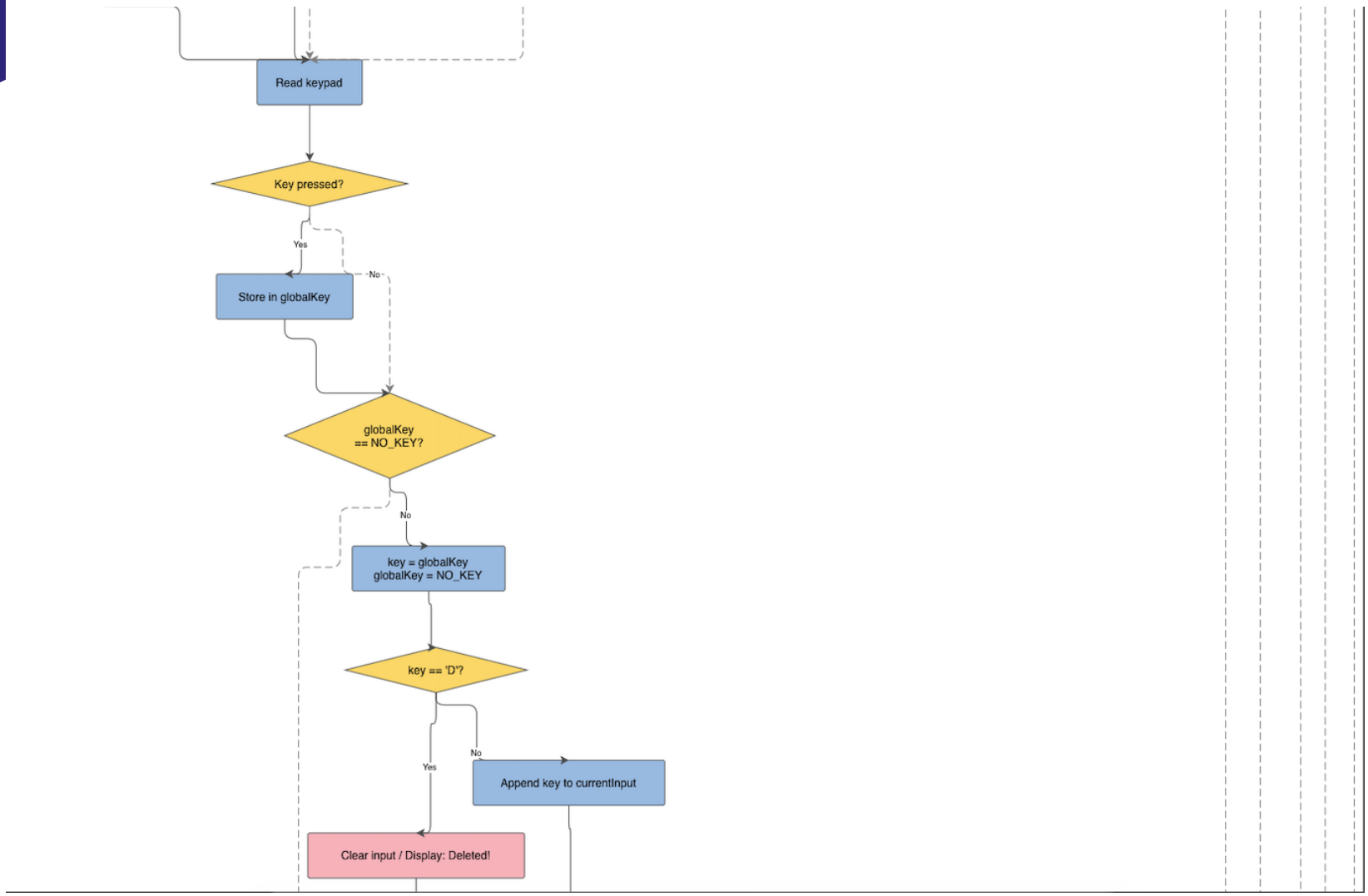

Color_Attempt(int Attempts, bool approved)
approved == true ?
No
no change since there is no way to deny the access with a single push button
Yes
RGB set to Green
Update NeoPixel Band
unsigned long timer = (millis()-startTime)/1000
level = 10 - (timer/12)
1 pixel of the 10x1 band is turned off each 10s
timer >= 120 ?
Yes
No
10x1 NeoPixel band cleared
LCD displays ("Time Over!") RGB in red
EndAttempt()

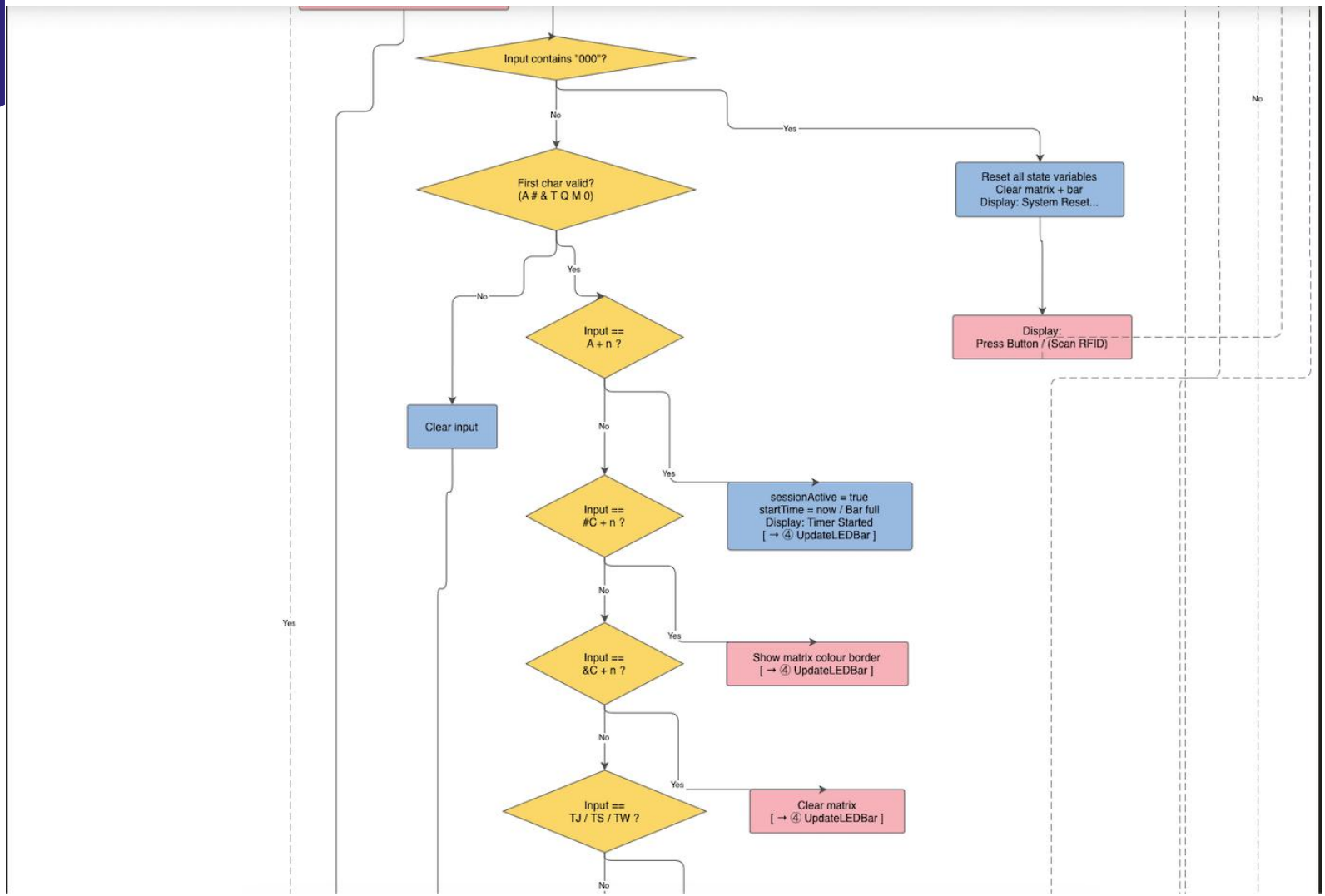

CompleteOrder()
ordersDelivered++
EndAttempt()

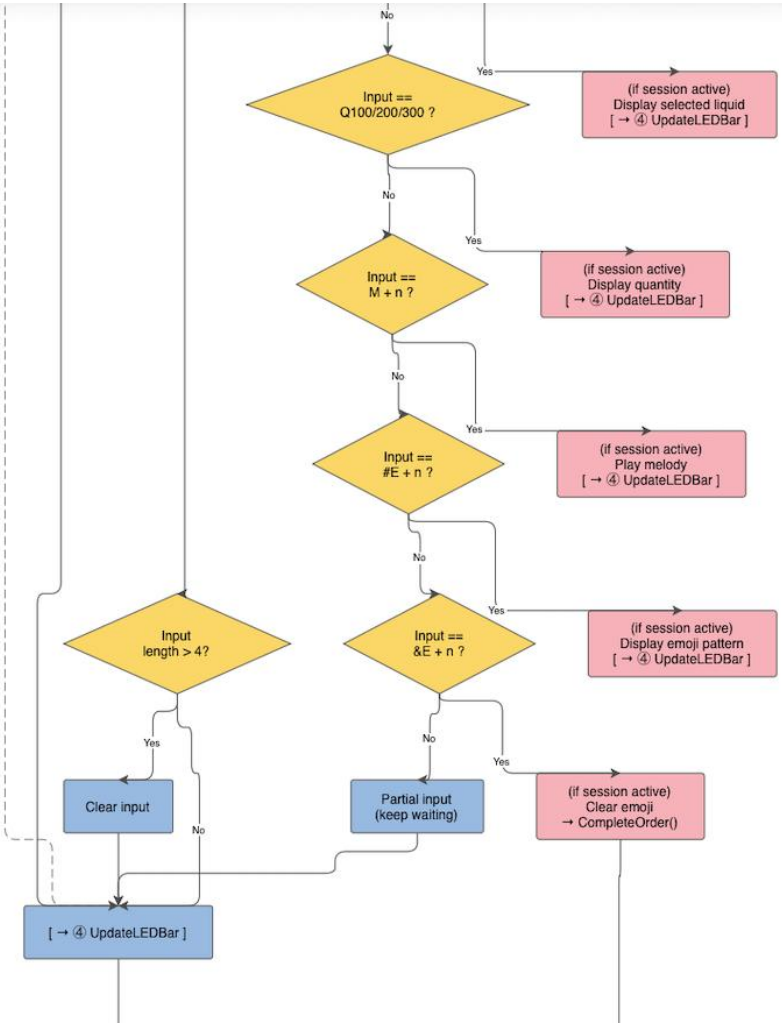

EndAttempt()
LCD displays ("RA:") + number of remaining attempts
LCD displays 3-totalAttempts
LCD displays the number of orders delivered
ordersDelivered ==3 ?
Yes
No
Displays "-MAX QUOTA"
10x1 NeoPixel Band cleared
sessionActive = false
push button reset to low state

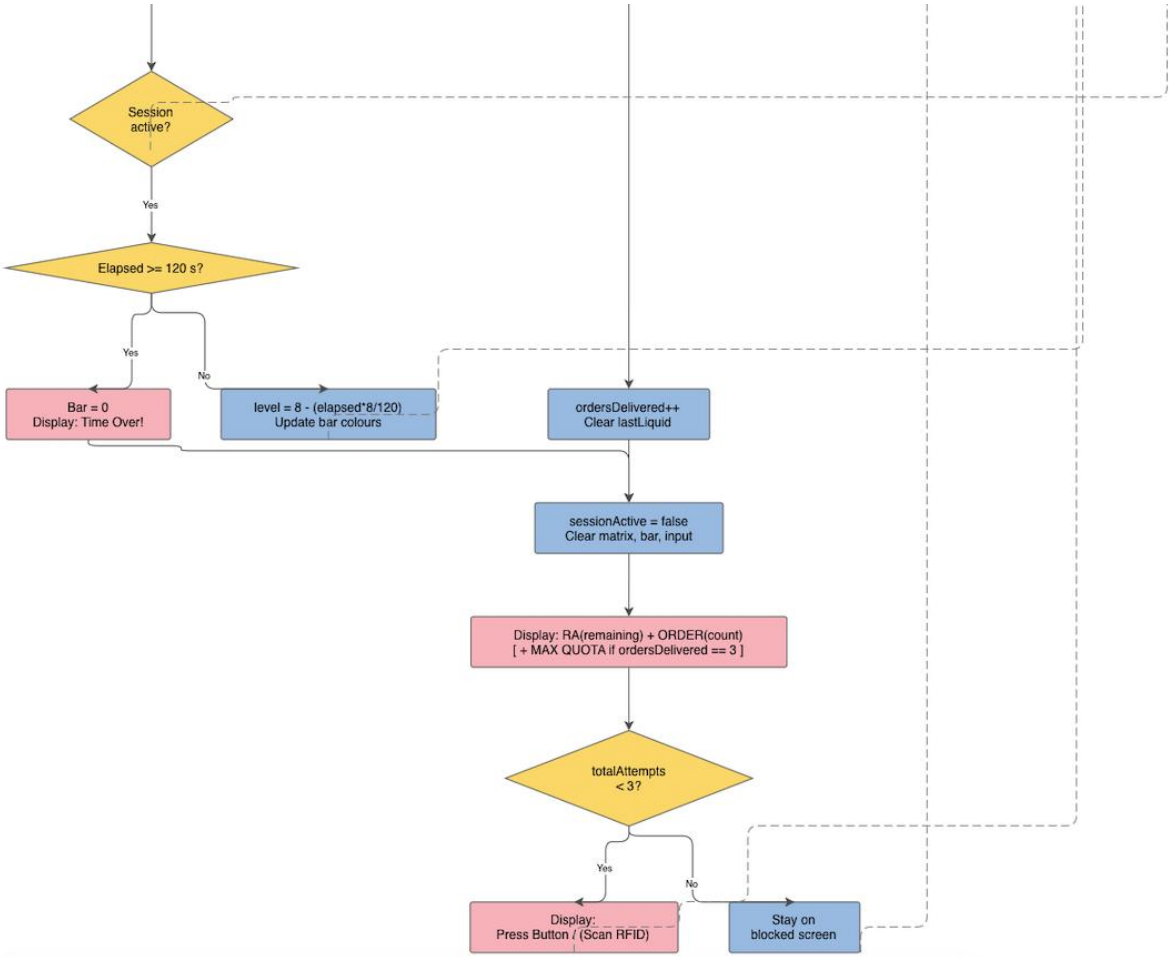
3.2. The Flowchart of the tinkercad part(The software side)


START
Init LCD
Init NeoPixel matrix (64 LEDs) + bar (8 LEDs)
Configure button as INPUT_PULLUP
Display: Press Button / (Scan RFID)
MAIN LOOP
Button HIGH
(not pressed)?
No
Debounce OK?
(>= 500 ms)
Yes
Yes
attempts >= 3?
No
No
Yes
totalAttempts++
accessApproved = true
Display: Approved Access
Display:
Blocked: RA0


Read keypad
Key pressed?
Yes
No
Store in globalKey
globalKey
== NO_KEY?
No
key = globalKey
globalKey = NO_KEY
key == 'D'?
Yes
No
Append key to currentInput
Clear input / Display: Deleted!


Input contains "000"?
No
Yes
First char valid?
(A # & T Q M 0)
Reset all state variables
Clear matrix + bar
Display: System Reset...
Yes
No
Input ==
A + n ?
Display:
Press Button / (Scan RFID)
Clear input
No
Yes
Input ==
#C + n ?
sessionActive = true
startTime = now / Bar full
Display: Timer Started
[→ ④ UpdateLEDBar]
No
Yes
Input ==
&C + n ?
Show matrix colour border
[→ ④ UpdateLEDBar]
No
Yes
Input ==
TJ / TS / TW ?
Clear matrix
[→ ④ UpdateLEDBar]
No
Yes
No


No
Input ==
Q100/200/300 ?
Yes
(if session active)
Display selected liquid
[→ ④ UpdateLEDBar]
No
Input ==
M + n ?
Yes
(if session active)
Display quantity
[→ ④ UpdateLEDBar]
No
Input ==
#E + n ?
Yes
(if session active)
Play melody
[→ ④ UpdateLEDBar]
No
Input ==
&E + n ?
Yes
(if session active)
Display emoji pattern
[→ ④ UpdateLEDBar]
No
Input
length > 4?
Yes
Clear input
No
Partial input
(keep waiting)
Yes
(if session active)
Clear emoji
→ CompleteOrder()
[→ ④ UpdateLEDBar]


Session active?
Yes
Elapsed >= 120 s?
Yes
No
Bar = 0
Display: Time Over!
level = 8 - (elapsed*8/120)
Update bar colours
ordersDelivered++
Clear lastLiquid
sessionActive = false
Clear matrix, bar, input
Display: RA(remaining) + ORDER(count)
[+ MAX QUOTA if ordersDelivered == 3]
totalAttempts
< 3?
Yes
No
Display:
Press Button / (Scan RFID)
Stay on
blocked screen

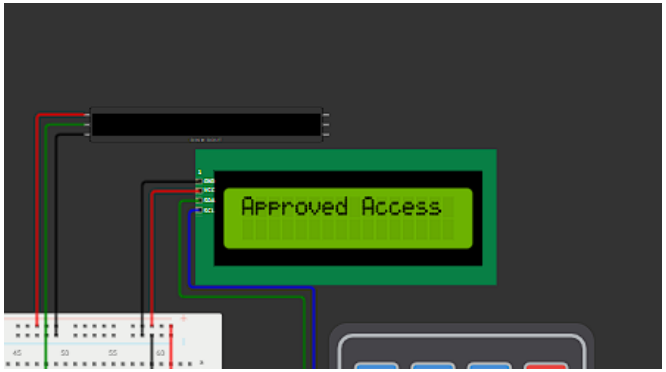
Software Adaptations for TinkerCAD

- Access Control: digitalRead(BUTTON_PIN)==LOW replaces RFID card/tag check
- LCD Colour Feedback: digitalWrite/analogWrite on external RGB LED (D13=R, D11=G, D12=B)
- Countdown Timer: an 8 LED strip instead of 10-LED Grove Bar
- Emoji Display: we replaced the 8x8 matrix 8 led strips of 8 leds each
- Melody: Same piezo buzzer and Arduino tones.

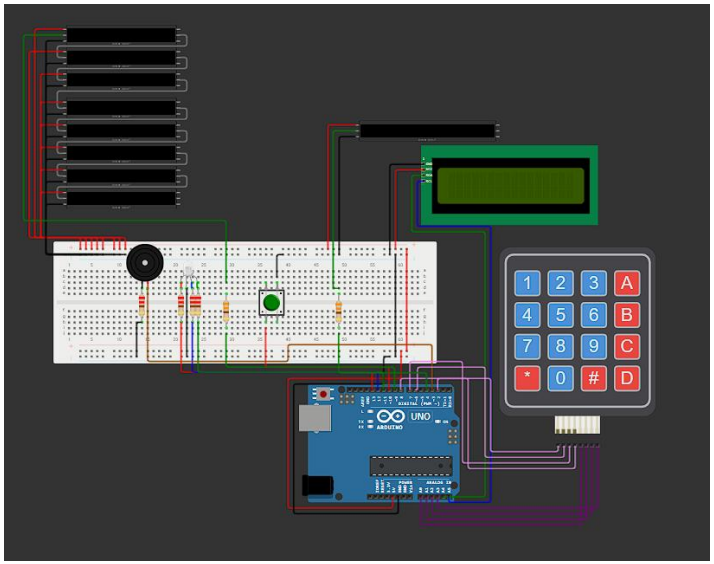
Constraints Respected with Adapted Components

- Same 3 attempts / 3 orders per day logic — all scenarios (A, B, C, D) implemented
- Same keypad layout and key mapping: A, T, Q, J, S, W, M, E, C, #, &, D
- Same liquid types: TJ (Juice), TS (Soda), TW (Water) and quantiies: Q100, Q200, Q500
- 3 different melodies (M1, M2, M3) and 3 distinct emojis with a certain command for an attmept (#E1, #E2, #E3)
- LCD still displays: access status, selected liquid, quantity, remaining attempts

3.3. Obtained Results



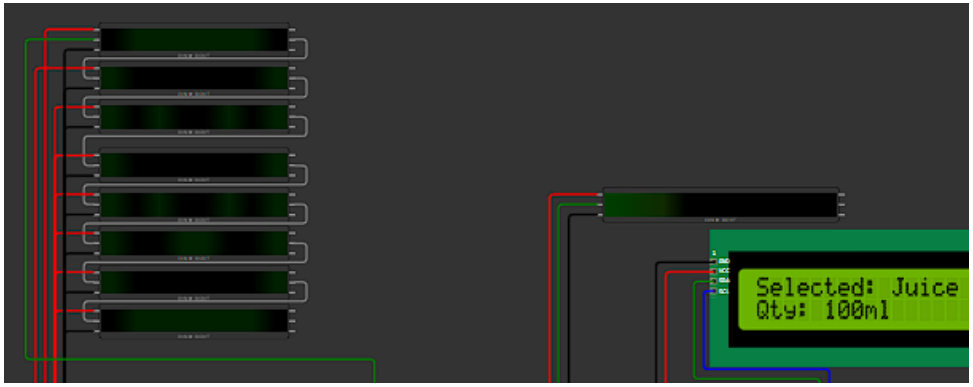
LCD "Approved Access" + RGB LED showing GREEN



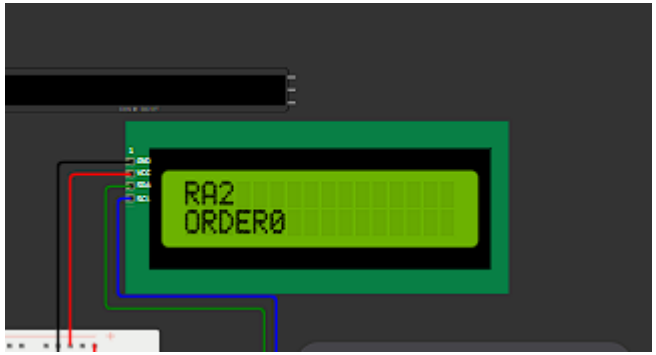
Wokwi circuit overview with all components



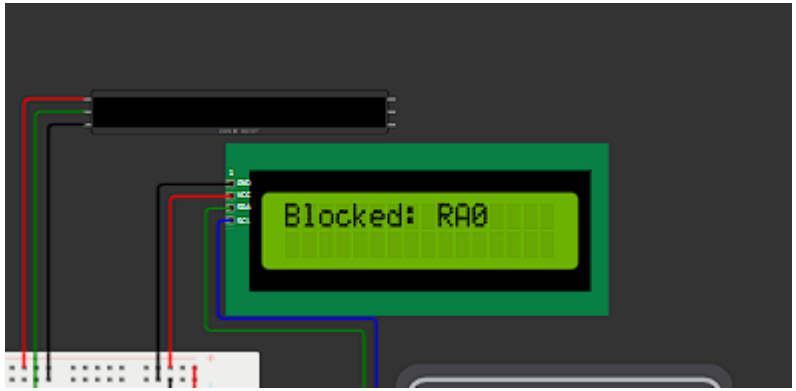
- *LCD "Selected: Juice" / "Quantity: 100 ml"*



Neopixels 10x1 band



LCD "RA2" / "ORDER0" at end of attempt



LCD “Blocked" + RGB LED Green

4. Comparison between the Practical & Tinkercad parts

Comparison between the Practical & Tinkercad parts

Materials Comparison

- RFID RC522 → Replaced by a Push Button: RFID not available in TinkerCAD
- Grove LCD RGB → I2C LCD + external RGB LED: Grove LCD library unavailable, the colour of the background cannot be directlty modified on tinkercad.
- Grove LED Bar (10 LEDs) → 10x1 Neopixel band: Grove LED Bar library unavailable
- Grove 8×8 LED Matrix → Combination of 8x1 NeoPixel bands: Custom emoji/LED patterns
- Speaker + MOSFET BS170 → Piezo Buzzer D3: Same tone functionality
- 4×4 Keypad: Identical layout and wiring in both versions

Software Differences

- Access: mfrc522 UID verification → digitalRead(BUTTON_PIN) == LOW
- LCD colour: lcd.setRGB(r,g,b) → RGB LED pins on d11, d12 and d13
- Countdown: bar.setLevel(n) → NeoPixel fill/setPixelColor patterns
- Emoji: matrix.displayFrames(frame,...) → NeoPixel fill/setPixelColor patterns

Common Specifications (both versions)

- Same 3 attempts/day and 3 orders/day logic — all scenarios (A, B, C, D) implemented
- Same 120-second countdown logic — adapted for available hardware
- Same keypad key mapping: A, T, Q, J, S, W, M, E, C, #, &, D
- Same 3 distinct melodies (M1, M2, M3) using tone() on D3
- Same void loop() structure: all logic in called functions, no instructions directly in loop()

5.Gantt Chart and project managment

Team Member Participation

- *Elio Saade: Circuit Design & Wiring, Practical Hardware Flowchart, Tinkercad Simulation Coding, Video editing*
- *Hugo Vrijhof: Practical Circuit Coding, Circuit Design, Gantt Chart, Team management*
- *Axel Estardié: Tinkercad and Wokwi Simulation Circuit Wiring, Practical Software Flowchart, Simulation Hardware Flowchart*

Project duration: May 4th, 2026 to May 19th, 2026
Tasks covered: circuit design, Arduino coding, Tinkercad simulation, troubleshooting, documentation, presentation preparation.

Liquid Vending Gantt Chart

TITRE DU PROJET	Liquid Vending	Organization	ECAM
Project members	Elio SAADE, Axel ESTARDIÉ, Hugo VRIJHOF	DATE	19/05/26

Number	Task name	Task manager	Initial date	Final date	Length	Overall task status	Phase One							Phase Two							Phase Three	
							Week 1							Week 2							Week 3	
							M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T
1	**Initial project workload organization**																					
1.1	Requirements analysis & task distribution		04/05/26	06/05/26	2	100 %																
1.2	Flowchart planning of complete processes		06/05/26	07/05/26	1	100 %																
2	**Part 1: Practical Circuit & Script**																					
2.1	Practical wiring & pin layout logic		07/05/26	08/05/26	1	100 %																
2.2	Coding step sequencing logic(Type select --> Qty select)		08/05/26	10/05/26	2	100 %																
2.3	Implementing M1-M3 music & #E/&E attempt-specific emojis		10/05/26	12/05/26	2	100 %																
3	**Part 2: Tinkercad Circuit & Script**																					
3.1	Component adaptation analysis (substituting unavailable practical parts)		12/05/26	13/05/26	1	100 %																
3.2	Tinkercad virtual schematic wiring & component connection		13/05/26	14/05/26	1	100 %																
3.3	Adapting script for Tinkercad component constraints		14/05/26	16/05/26	2	100 %																
4	**Project deliverables**																					
4.1	Project PowerPoint & circuit flowcharts		16/05/26	17/05/26	1	100 %																
4.2	Live presentation preparation & script rehearsal		17/05/26	18/05/26	1	100 %																
4.3	Automated compilation checks & code formatting		18/05/26	19/05/26	1	100 %																
4.4	Final Project Submission		19/05/26	19/05/26	0	100 %																

6. Conclusion

Conclusion

•**Successful Implementation:** Designed and built a fully functional liquid vending machine using the RFID RC522 module.
•**Dual-Platform Success:** Achieved complete hardware functionality in both the physical prototype and the Tinkercad simulation.
•**Core Constraints Met:** Successfully enforced a 120-second order countdown, a 3-attempt security limit, and a maximum of 3 daily orders.
•**Custom User Interface:** Programmed a unique keypad map alongside 3 distinct emojis and melodies that dynamically change based on the active attempt.
•**Order Control Features:** Integrated functional "Reset" and "Delete" commands, allowing users to wipe a current entry or completely restore system attempts.

Key skills reinforced

- Embedded C++ programming and Arduino IDE
- SPI and I2C communication protocols
- Electronic circuit design and breadboard wiring
- State machine design for multi-step user interactions
- Component substitution and hardware-software co-design

7. Feedback

Difficulties Encountered

•**Tinkercad Obstacles:** Faced issues finding and replacing missing components, and the platform ultimately failed to run the code.

•**Learning Curve:** Used AI to learn how to configure and script functions for the unfamiliar hardware devices.

•**Wokwi Solution:** Switched to Wokwi, where the identical code successfully executed without errors.

•**Easy Hardware Assembly:** Encountered no issues building the physical circuit since the team had previously used all the components.

Learning Outcomes Gained

- *Efficient Substitution Solutions for Practical elements on simulation softwares*
- *Improved understanding of overall SPI and I2C protocols, and RFID module.*
- *Improvement of coding knowledge*

General Feedback on the Project

•**Engaging & Educational:** The project felt like the continuity of the lab sessions, successfully combining familiar concepts with other elements.

•**Strong Teamwork:** It was considered the one of the best project of the year, providing a challenge that required excellent team work to finish on time.

•**Future Inspiration:** Seeing the potential of Arduino inspired a desire to learn more and experiment with projects at home.

8.DEMO of the practical part


YOUNGLA

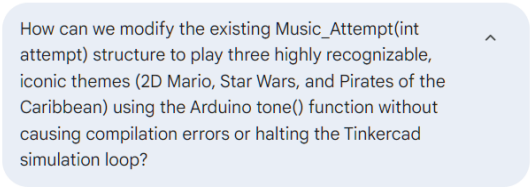
9.References and IA use

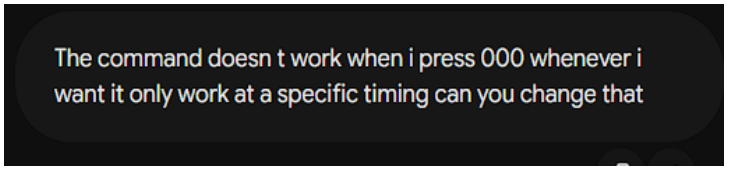
References Used

- Arduino Official Documentation: https://docs.arduino.cc
- ECAM LaSalle LAB1, LAB2, LAB3 course materials (Moodle)
-

AI Tools Used

• Claude, for code debugging assistance and programation of new components such as the Neopixels.
• Gemini, for melody creation
Important to note ai was strictly only used for commands we had not seen before.
• *AI tools were used in approximately 12% of the project which corresponds to only a minor part of the codes.*





Thank you
for your attention



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